



Integrating artificial intelligence in biodiversity conservation: bridging classical and modern approaches

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Abstract

Preserving biodiversity is crucial for maintaining ecological balance; however, traditional conservation methods often face various limitations. In most cases, the efficacy of these approaches is frequently constrained by difficulties in scaling and the absence of up-to-date data, hence requiring the incorporation of novel technology. The latest progress in the field of artificial intelligence (AI) offer transformative potential for enhancing contemporary conservation endeavors. There is a growing utilization of AI technologies, like machine learning and data analytics, to improve species identification, habitat monitoring, and threat assessment with exceptional precision and effectiveness. This study explores how AI is incorporated to enhance conventional conservation methods, particularly in the areas such as data analysis, species identification, and habitat monitoring. This paper examines a number of case studies that demonstrate the successful use of AI, with a particular focus on notable advancements in data management, predictive modeling, and resource allocation. The findings highlighted the significance of synergistic methodology that integrates the strength of traditional techniques with the flexibility of contemporary technologies, hence facilitating the development of more resilient conservation solutions. This study also discusses the potential implications for future research and the practical use of AI in the field of conservation. It highlights the strategy of seamless integration to justify both scientific investigation and conservation results.

Keywords Biodiversity conservation · Artificial intelligence · Machine learning · Ecosystem management · Predictive modeling · Conservation technologies

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Introduction

Biodiversity conservation is crucial for maintaining natural equilibrium and promoting human welfare (Rawat and Agarwal 2015). The preservation of biodiversity ensure the long-term viability of ecosystem, providing essential ecological, social, and economic benefits(Niesenbaum 2019). These services encompass furnishing essential resources i.e., food, water, and medicine; regulatory functions like disease management and climate regulation; supporting functions such as soil formation, nutrient cycling, and cultural functions (Naeem et al. 2016).

Traditionally, biodiversity conservation has relied on approaches such as creating protected areas, relocating endangered species, and implementing legal safeguards to protect habitats (Heywood 2019). However, these conventional methods face significant challenges in today's rapidly changing global landscape, characterized by ongoing difficulties such as habitat degradation, climatic variability, and the proliferation of non-native species (Richardson and Whittaker 2010). In response to these challenges, recent approaches have begun integrating advanced technologies such as AI, remote sensing, and genetic analysis to enhance conservation efforts. As natural and human landscapes change, the efficacy of these traditional methods is being increasingly scrutinized (Redclift and Springett 2015).

Contemporary methodologies, particularly those applying AI for tasks like data analysis and predictive modeling, present auspicious prospects for enhancing biodiversity conservation (Obaideen et al. 2024). AI possesses the capability to efficiently and accurately analyze extensive quantities of data, enabling tasks such as species identification through image recognition and habitat monitoring via satellite imagery (Ditria et al. 2022). This technological transition serves as a crucial connection between traditional conservation techniques and modern challenges, enabling the implementation of more adaptable, flexible, and efficient conservation strategies (Bibri et al. 2024).

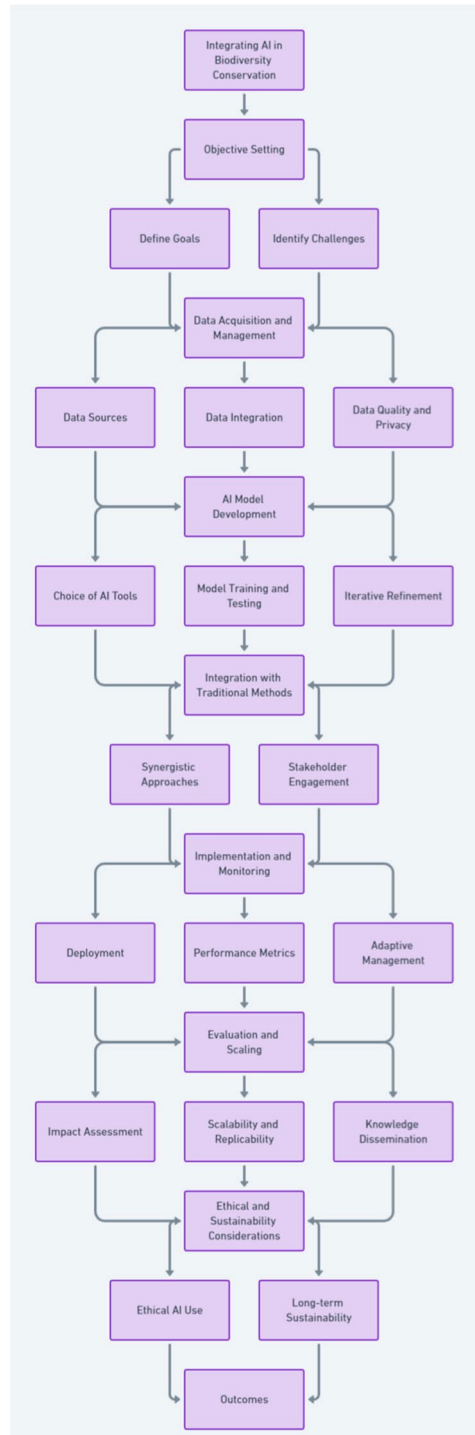
This research is justified by the urgent need to enhance biodiversity conservation efforts, as traditional methods alone are insufficient to address the escalating threats posed by global change and human activity. The primary objectives of this study are to: (1) evaluate how AI can enhance traditional conservation methods, (2) analyze case studies demonstrating successful AI applications in biodiversity conservation, and (3) propose a framework for integrating AI technologies into existing conservation strategies. AI is a potent instrument for swift, data-centric observations that can result in more efficient conservation measures. The aforementioned viewpoint holds significant importance in the conservation of biodiversity, which subsequently upholds the provision of vital ecological services necessary for human life and the overall well-being of the planet. By achieving these objectives, this article aims to present a revolutionary approach to environmental stewardship that integrates conventional conservation strategies with modern computational methodologies, ultimately leading to more effective and resilient conservation solutions (Fig. 1). The combination of traditional and contemporary approaches can enhance the ability to make well-informed and proactive decisions regarding conservation, so establishing a higher benchmark for environmental study and action.

Classical approaches to biodiversity conservation

Discussion on traditional conservation strategies

Historically, conventional conservation measures have formed the foundation for protecting biodiversity, such as the establishment of national parks like Yellowstone in the USA, which

Fig. 1 Integrating AI in biodiversity conservation framework



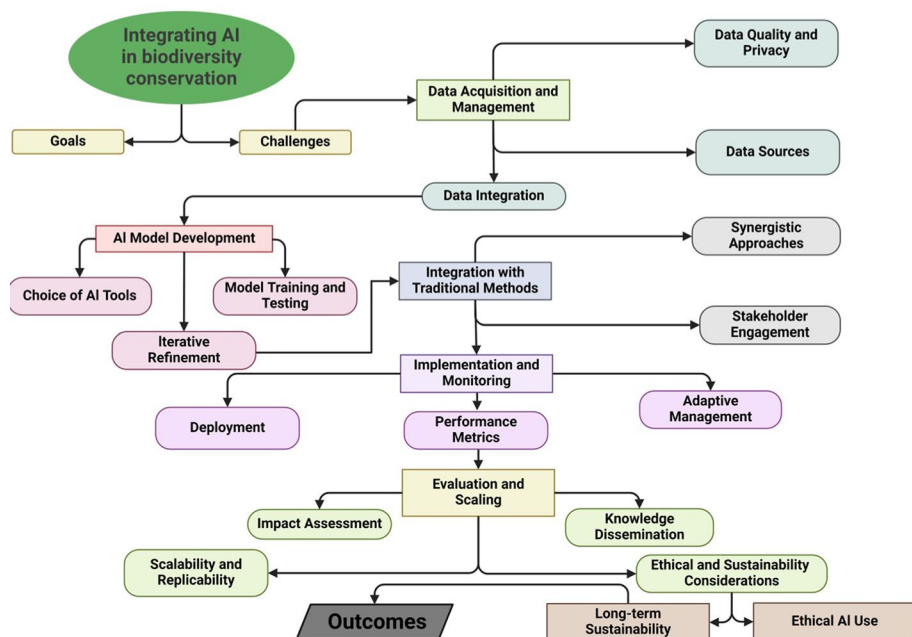


Fig. 1 (continued)

has successfully preserved numerous species (Robinson 2006). The primary emphasis of these techniques lies in the conservation of habitats, the creation of protected areas, and interventions tailored to specific species (Watson et al. 2011). Although traditional conservation methods such as the creation of protected areas, species relocation, and legal protections have shown favorable results in various contexts, their execution frequently encounters substantial obstacles, such as financial limitations, clashes with indigenous communities, and restricted expandability (Lister 1998). To enhance the success of these conservation measures, it is crucial to strengthen policy enforcement and engage the community (Fig. 2).

Habitat preservation

Traditional habitat preservation efforts often involve establishing reserves and implementing restoration projects aimed at maintaining ecosystems (Lister 1998). For instance, the restoration efforts in the Atlantic forest of Brazil have been instrumental in recovering native species and restoring ecological functions (Hoffmann 2010). This technique serves the dual purpose of safeguarding biological diversity and maintaining the ecological functions, thus playing a vital role in the long-term survival of species (Moonen and Bàrberi 2008). However, habitat conservation faces significant challenges, including habitat fragmentation due to urban development, agricultural expansion, and climate change impacts, which can disrupt ecological connectivity and reduce habitat quality (Hoffmann 2022; Ssenku et al. 2022). Given that habitat fragmentation is primarily driven by human activities such as urban development, agriculture, and climate change, habitat conservation alone

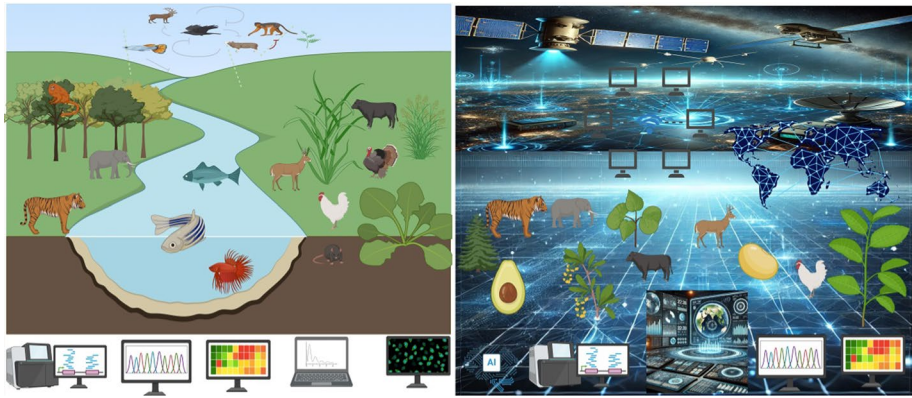


Fig. 2 Left Side: Illustration of a natural ecosystem with diverse species monitored using traditional tools like genetic sequencing and environmental assessments. Right Side: AI-driven conservation using satellites, global monitoring, and data analytics to track and protect biodiversity worldwide

may not suffice to address these multifaceted challenges. Therefore, it is essential to integrate habitat conservation with complementary strategies that target these external pressures. These complementary efforts may include community engagement initiatives that promote sustainable land use practices, restoration projects that reconnect fragmented habitats, and policy advocacy aimed at reducing human impacts on ecosystems.

Legal frameworks and policies

Legal frameworks and policies play a crucial role in establishing the essential foundation for the enforcement of conservation initiatives (Cooney 2004). Examples include the Endangered Species Act (ESA) in the U.S.A (Act 1973) and the Convention on Biological Diversity (CBD), which aim to protect endangered species and their habitats. The Endangered Species Act in the United States has led to the recovery of species like the bald eagle and gray wolf through stringent legal protections (Cooney 2006). The purpose of these regulations is to safeguard endangered species and habitats, govern detrimental actions, and promote the sustainable utilization of resources (Berrens 2001). Although legal frameworks have demonstrated some effectiveness, they must undergo continuous evolution in order to effectively tackle emerging environmental concerns and integrate scientific discoveries and community rights into conservation activities (Techera 2013b). Studies have shown that populations of listed species under the ESA have increased significantly due to targeted recovery plans and habitat restoration efforts (Act 1973; Evansen et al. 2021). These frameworks must include recent AI based approaches to effectively tackle emerging environmental concerns and integrate scientific discoveries and community rights into conservation activities (Techera 2013a).

Community-based conservation

Community-based conservation entails the active participation of local populations in the administration and safeguarding of natural resources, acknowledging that local administration can result in more enduring conservation results. An example is the CAMPFIRE program in Zimbabwe, which empowers local communities to manage wildlife resources, leading to increased wildlife populations and community benefits (Pimbert and Pretty 1997). This methodology utilizes conventional knowledge and customs, cultivating a perception of possession and accountability among individuals within the group (Shackeroff and Campbell 2007). For instance, local fishing practices that incorporate seasonal cycles and sustainable harvesting methods reflect conventional knowledge that can enhance fish populations and ecosystem health. Similarly, indigenous land management techniques, such as controlled burns, have been shown to improve habitat conditions for various species. Nevertheless, in order to achieve optimal effectiveness, it necessitates sufficient assistance in terms of enhancing capabilities and ensuring fair distribution of benefits.

Strengths and limitations of classical approaches

Traditional conservation approaches have notable strengths due to their historical use and substantial empirical evidence supporting their effectiveness. While these methods have successfully prevented species extinction, they often struggle with scalability, as seen in overexploited fisheries where traditional regulations failed to adapt the increasing demand (Agrawal and Redford 2006). These approaches primarily focus on habitat preservation, regulatory measures, and targeted interventions designed to protect specific species. However, these methods often face significant limitations. For instance, protected areas may not adequately account for shifting species distributions caused by climate change, leading to ineffective conservation outcomes (Richardson and Whittaker 2010). Additionally, traditional methods may lack flexibility and responsiveness; this has been evident in cases where invasive species management has been slow to adapt to new threats (Moonen and Bärberi 2008). They have played a critical role in preventing the extinction of several species and safeguarding essential ecological systems. Nevertheless, these approaches are frequently restricted by their responsiveness rather than proactiveness, budgetary and logistical limitations, and occasionally insufficient incorporation with contemporary scientific methodologies and community requirements (Moonen and Bärberi 2008; Ibisch et al. 2010). The aforementioned constraints underscore the imperative of incorporating novel, adaptable, and all-encompassing methodologies, such as those offered by nascent technologies like AI.

Modern approaches including AI technologies in biodiversity conservation

Introduction to modern technologies in conservation

Contemporary technologies, especially AI, are significantly transforming biodiversity conservation by enhancing data analysis capabilities and enabling real-time monitoring. Technologies such as satellite imagery for land-use monitoring and AI-driven analytics for species

identification are revolutionizing conservation practices (Ritts and Bakker 2021). Modern conservation techniques heavily rely on various tools and technologies, including genetic analysis methods, remote sensing, AI, and autonomous systems (Ditria et al. 2022; Bibri et al. 2024). These technologies provide valuable insights and efficiency that were previously unachievable. Such technologies facilitate the ability of conservationists to forecast alterations, simulate probable consequences, and proactively execute strategies to save biodiversity (Fig. 2).

Genetic technologies

Genetic technologies play a vital role in contemporary conservation efforts by offering techniques to evaluate biodiversity, comprehend population genetics, and monitor the genetic well-being of species. Methods such as DNA barcoding and environmental DNA (eDNA) analysis enable the recognition of species based on small genetic remnants present in the environment, providing a non-intrusive approach to monitor biodiversity and the well-being of the ecosystem (Cristescu and Hebert 2018; Antonelli et al. 2021; Piaggio 2021; Silvestro et al. 2022). These methodologies are particularly valuable for identifying rare or elusive species and assessing the impacts of environmental changes on genetic diversity. By enabling precise species identification and monitoring genetic health, these technologies support conservation strategies aimed at preserving genetic diversity and informing management decisions. AI algorithms can analyze genetic data more efficiently, allowing for quicker assessments of biodiversity (Cristescu and Hebert 2018). These methodologies hold significant value in the identification of uncommon or enigmatic species, as well as in the evaluation of the effects of environmental alterations on genetic variability.

Remote sensing and monitoring

Satellite images and airborne drones are examples of remote sensing technologies that offer extensive capabilities for monitoring habitats, land-use changes, and species distributions over vast regions. These tools capture high-resolution images that can reveal changes in vegetation cover, land use, and habitat fragmentation over time (Lahoz-Monfort and Magrath 2021). This ongoing monitoring capability allows conservationists to respond quickly to threats such as deforestation, wildfires, and illegal wildlife activities, thereby enhancing the effectiveness of conservation efforts. Remote sensing data can be processed using AI to identify habitat changes with greater accuracy (Lahoz-Monfort and Magrath 2021). These technological advancements enable the ongoing monitoring of environmental fluctuations, hence facilitating prompt reactions to various challenges such as deforestation, wildfires, and illicit wildlife operations (Jiménez López and Mulero-Pázmány 2019; Buchelt et al. 2024). State-of-the-art sensors and imaging have the capability to identify alterations at the micro-habitat level, which is essential for implementing focused conservation measures (Simaika et al. 2024).

Data-driven conservation planning

Data-driven conservation planning leverages extensive datasets derived from many sources, including remote sensing, field observations, and genetic data (Shaikh et al. 2022). These methods analyze spatial patterns and relationships within the data to inform strategic decisions (Shaikh et al. 2022). By forecasting the most effective locations for conservation measures and evaluating the potential results of various management systems, this

methodology allows for the optimization of resource allocation (Xiang et al. 2021). This ensures that conservation efforts are both efficient and impactful.

Exploration of AI in conservation

The integration of deep learning for species identification, computer vision for habitat assessment, and predictive analytics for forecasting species migration into ecological research and management results in the transformation of conservation methods through the utilization of AI. For instance, AI algorithms have been employed to analyze camera trap images for species identification in projects like Snapshot Serengeti, leading to improved data collection efficiency (Lamba et al. 2019). AI systems have the capability to handle and examine extensive amounts of environmental data, ranging from identifying species in camera trap images to detecting patterns in temporal data (Lamba et al. 2019; Green et al. 2020). This enables the making of conservation decisions in real-time (Ditria et al. 2022). AI has been employed to automate the identification and quantification of species, assess the condition of habitats, and forecast the migration patterns of animals across various terrains, thereby furnishing conservationists with a versatile set of tools.

Machine learning models for species identification and behavior prediction

The AI models have demonstrated notable efficacy in the domains of species identification and behavior prediction (Merabet et al. 2021). Through the examination of data derived from many sources, including audio recordings and field-captured images, these models possess the capability to effectively and precisely classify species, even within intricate ecological habitats (Zhang et al. 2023). In addition, predictive models have the ability to forecast the behaviors and movements of species, which can assist in the management of conflicts between humans and wildlife, as well as in the creation of wildlife corridors that improve connection between environments that are fragmented (Nandutu et al. 2022).

AI-driven data analytics for habitat monitoring and threat assessment

AI-driven data analytics transform habitat monitoring and threat assessment by combining several data streams to offer a comprehensive perspective on the health of the ecosystem. The aforementioned techniques has the capability to accurately examine alterations in vegetation patterns, water quality, and land use, hence providing valuable insights into the effects of human-induced stresses on natural environments (Chowdhury et al. 2024). AI models play a crucial role in forecasting forthcoming environmental conditions and evaluating prospective hazards, hence facilitating proactive conservation efforts (Isabelle and Westerlund 2022; Akter 2024).

Autonomous systems for invasive species control

Autonomous systems, such as unmanned aerial vehicles (UAVs) and robotic devices, are utilized for the purpose of managing invasive species, which pose a significant risk to biodiversity. These systems are capable of functioning in difficult conditions to eliminate or control exotic species, administer specific pesticides, or revive indigenous plants (Gonzalez et al. 2016; Martinez et al. 2020). The incorporation of AI allows these systems to

effectively detect and focus on particular species, resulting in decreased labor expenses and mitigated human influence on delicate ecological systems (Galaz et al. 2021).

Case studies highlighting AI in conservation

The integration of AI into conservation efforts has led to significant advancements in monitoring endangered species (Amur leopards in Russia) and managing habitats effectively (Rozhnov et al. 2019). The video traps were strategically placed by researchers inside the habitat of leopards in order to facilitate the automated identification of individual animals based on their distinct coat patterns (Alibhai et al. 2023). By using AI algorithms that have been trained on a vast collection of photos, it is possible to differentiate leopards from other animals and effectively track their population and overall well-being. This technology enables uninterrupted, instantaneous monitoring without the need for human involvement, resulting in a substantial decrease in the labor and expenses typically linked to such endeavors. Consequently, it plays a vital role in updating population estimates and assisting conservation strategies. Consequently, it plays a vital role in updating population estimates and assisting conservation strategies, as demonstrated by studies showing improved population monitoring and management outcomes. For example, research highlights how AI-driven camera trap systems have enabled real-time tracking of Amur leopards, leading to more accurate population estimates and enhanced conservation actions (Alibhai et al. 2023). Similarly, a study found that AI analysis of satellite imagery significantly improved the detection of deforestation activities in the Amazon, allowing for timely interventions that contributed to better forest conservation and species protection (Casas and Torres 2023). Furthermore, the cost-effectiveness of AI technologies allows for savings that can be reinvested into further conservation efforts. For instance, the automation of data collection processes reduces labor costs, enabling funds to be redirected toward habitat restoration projects or community engagement initiatives (Albuquerque et al. 2024). AI has the capacity to significantly improve project efficiency, successfully handle extensive information, and provide timely insights crucial for preserving species and ecosystems. The effective use of AI technology in these situations highlights their usefulness in tackling intricate conservation issues across various ecological environments.

AI has been used for ecological management in the Amazon rainforest. Conservationists used AI to examine extensive satellite footage in order to promptly identify deforestation operations, alterations in land utilization, and disruptions. The training of machine learning models aimed to distinguish between natural and anthropogenic alterations occurring within the ecological system (Casas et al. 2023). The utilization of AI technology has facilitated proactive management strategies by accurately forecasting deforestation patterns and promptly detecting illicit activities, surpassing conventional approaches. This has played a crucial role in influencing policy choices and allocating resources for conservation endeavors (Albuquerque et al. 2022).

Moreover, AI has revolutionized marine conservation efforts in the Coral Triangle region of Southeast Asia (Hoeksema 2007). AI was used by scientists to assess and map coral reefs using photographs captured by underwater drones (McLeod et al. 2010). The AI was assigned the responsibility of detecting coral health indicators, such as instances of bleaching, and evaluating the dynamics of fish populations. The use of this program has facilitated a much expedited and precise evaluation of reef well-being in comparison to conventional techniques, hence enabling prompt treatments (Clifton & Foale 2017). In

addition, the use of AI insights is employed in the optimization of conservation results by designing marine protected areas that are more successful.

AI has the capacity to significantly improve project efficiency, successfully handle extensive information, and provide timely insights that are crucial for the preservation of species and ecosystems. The effective use of AI technology in these situations highlights their usefulness in tackling intricate conservation issues in many ecological environments.

Discussion

Comparative analysis of classical and modern approaches

An examination of classical and modern approaches to biodiversity conservation highlights the unique advantages and difficulties associated with each paradigm. While traditional methods have been effective in creating protected areas, they often lack flexibility; modern technologies like AI offer adaptive solutions that can quickly respond to environmental changes (Anderson and Jenkins 2006). Traditional methods for addressing these challenges include establishing protected areas and implementing restoration projects aimed at reconnecting fragmented habitats. For instance, wildlife corridors have been created to facilitate species movement between isolated habitats, thereby enhancing genetic diversity and resilience (Noss et al. 2015). The utilization of these techniques has played a crucial role in regions such as the Galapagos Islands, where rigorous conservation efforts have effectively preserved distinct species (Muñoz-Viñas 2012). Despite these conventional strategies, limitations remain in their scalability and adaptability to rapidly changing environments. Table 1 explain the comparative analysis of traditional and AI based conservation.

In contrast, contemporary methods utilize technology progress such as genetic technologies, AI, and remote sensing to overcome these constraints (Marvin et al. 2016; Kerry et al. 2022; Singh et al. 2022). These tools provide flexible, expandable, and accurate approaches for conservation, easily adjusting to the intricacies of worldwide environmental issues (Shwartz et al. 2014). AI-powered models have revolutionized conservation tactics by offering real-time data and predicted insights for habitat and species monitoring, hence improving decision-making processes. AI has been employed in the Amazon jungle to forecast deforestation patterns and detect illicit logging activities, surpassing the efficiency of traditional methods (Raihan 2023). AI technologies can significantly enhance habitat preservation efforts by providing real-time data analytics for monitoring habitat conditions and predicting changes based on environmental variables. For example, machine learning models can analyze satellite imagery to identify areas at risk of degradation and prioritize them for conservation interventions (Ditria et al. 2022).

The incorporation of contemporary technologies in the field of conservation also facilitates the utilization of extensive data analysis capabilities that exceed conventional approaches (Pimm et al. 2015). Initiatives like as eBird utilize citizen science and AI to collect and analyze worldwide bird observation data, providing valuable information on migration patterns and population dynamics that guide conservation efforts at both local and global levels (Kelling et al. 2012; Johnson et al. 2020). The aforementioned demonstrates the potential of contemporary methodologies to supplement traditional techniques, hence augmenting their scope and effectiveness (Di Marco et al. 2017).

Table 1 Comparative analysis of traditional and AI-enhanced approaches to biodiversity conservation

Study title	Conventional methods used	Limitations of traditional methods	AI methods used	Outcomes with AI
Improving biodiversity protection through artificial intelligence	Traditional methods for biodiversity conservation often include random selection strategies based on species richness and simple monitoring techniques (Hoffmann 2022)	These methods frequently fail to account for the trade-offs between costs and benefits of conservation efforts, leading to inefficient allocation of resources and insufficient protection of species at risk (Hoffmann 2022)	AI methods applied include machine learning for wildlife monitoring, predictive modeling for species distribution, and automated data analysis from sensors and cameras. Additionally, AI supports the optimization of conservation strategies through data analysis and decision-making models (Dauvergne 2020)	The use of AI in conservation has led to improved monitoring of ecosystems, enhanced data analysis for decision-making, more efficient resource allocation, and better engagement with local communities for conservation practices. These outcomes collectively contribute to more effective biodiversity protection efforts (Dauvergne 2020)
Improving biodiversity protection through artificial intelligence	Traditional methods for biodiversity conservation often include random selection strategies based on species richness and simple monitoring techniques (Silvestro et al. 2022)	These methods frequently fail to account for the trade-offs between costs and benefits of conservation efforts, leading to inefficient allocation of resources and insufficient protection of species at risk (Silvestro et al. 2022)	The study employs the CAPTAIN (Conservation Area Prioritization Through Artificial Intelligence) framework, which uses reinforcement learning to optimize conservation planning by quantifying trade-offs in biodiversity protection within budget constraints (Silvestro et al. 2022)	The AI methods resulted in the protection of significantly more species from extinction compared to traditional methods, with algorithms demonstrating a 54% increase in the number of protected areas under the same budget, achieving more efficient conservation outcomes overall (Silvestro et al. 2022)
AI in Wildlife Conservation: A Comprehensive Overview	Traditional monitoring methods include manual tracking, camera traps, and visual observation, which require significant human involvement and time (Tuia et al. 2022)	These methods are time-consuming, can be inaccurate, and often lack real-time data collection, which limits their effectiveness in rapidly changing environments (Tuia et al. 2022)	AI methods used include machine learning for processing camera trap images, image recognition for species identification, and automated tracking systems for real-time data collection (Nagini et al. 2023)	Positive outcomes achieved through AI include improved accuracy in species monitoring, reduction in manual labor, faster data analysis, and enhanced ability to predict and mitigate threats to wildlife (Nagini et al. 2023)

Table 1 (continued)

Study title	Conventional methods used	Limitations of traditional methods	AI methods used	Outcomes with AI
Conservation-oriented restoration—a two for one method to restore	The study discusses the limitations of traditional conservation methods, which primarily rely on the designation of protected areas, assessments of biodiversity, and ex situ conservation without addressing the need for habitat modifications to effectively maintain ecosystem dynamics or react to alterations caused by anthropogenic pressures such as climate change and habitat degradation. Traditional methods typically do not involve active interventions that promote biodiversity recovery and ecosystem health (Volis 2019)	Traditional conservation methods often fail due to assumptions that natural recovery in protected areas can occur without human intervention. They inadequately address habitat degradation, which prevents natural processes necessary for biodiversity recovery. Moreover, the dependence on legal protection alone does not mitigate ongoing disturbances in the ecosystem, leading to the potential extinction of threatened species, as many species have limited dispersal capabilities and may not regenerate effectively without additional interventions. There is a clear need for adaptive management and more interactive approaches during restoration (Volis 2019)	AI techniques, particularly through reinforcement learning (RL), were applied to optimize conservation policies by assessing biodiversity over time and adapting strategies to meet conservation objectives. The CAPTAIN framework is a key innovation which integrates AI to enable real-time prediction and management of biodiversity, optimize data collection, and improve resource allocation in restoration planning. Various types of data monitoring techniques using AI are discussed, including automated image analysis and remote sensing for efficient ecosystem data retrieval (Ditria et al. 2022)	The implementation of AI in conservation practices has demonstrated a significant increase in the efficiency of biodiversity protection and management outcomes, leading to more effective restoration practices. The CAPTAIN framework, when tested, showcased a capacity to protect a higher percentage of species from extinction compared to traditional planning methods, including Marxan. Additionally, AI-assisted strategies facilitated cost-effective and adaptive management approaches, ultimately leading to improved ecosystem health through enhanced real-time monitoring and responsive management actions (Silvestro et al. 2022)

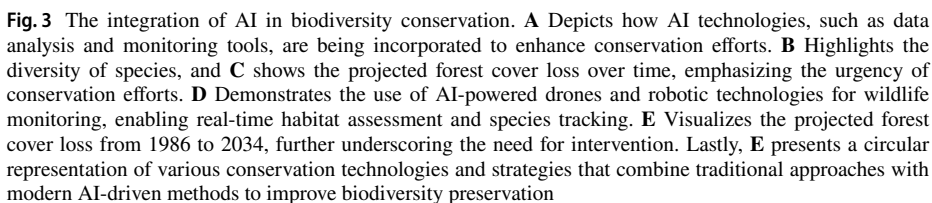
Moreover, contemporary conservation methods can more effectively address the socio-economic dimensions of biodiversity preservation (De Groot et al. 2012). Predictive modeling and real-time monitoring are utilized to facilitate more sophisticated management of human-wildlife interactions, thereby mitigating the constraints associated with traditional approaches that frequently encounter difficulties in community engagement and socio-economic integration (Du et al. 2023). AI models employed in wildlife corridors in Africa play a crucial role in forecasting and alleviating human-elephant conflicts, a task that cannot be accomplished only by static protected zones (Morrison 2019). While AI has shown great promise in enhancing biodiversity conservation efforts, several emerging challenges must be addressed to ensure its effective application in the future. As AI systems rely heavily on vast amounts of data, concerns regarding data privacy and security are paramount. Ensuring that sensitive ecological data is protected from misuse is critical. AI algorithms can inherit biases present in training data, which may lead to skewed results and misinformed conservation strategies. Addressing algorithmic bias is essential for equitable conservation outcomes. The implementation of AI technologies often requires significant financial and technical resources, which may not be available in all regions, particularly in developing countries. Ensuring equitable access to these technologies is vital.

Synergies between traditional methods and AI technologies

The potential synergy between the integration of AI technologies with traditional conservation methodologies offers a significant opportunity to increase biodiversity conservation by capitalizing on the respective strengths of both approaches (White et al. 2021). The convergence of AI's analytical powers with the grounded, locality-specific actions of classical methods is exemplified by a range of real-time applications. An illustrative instance is the utilization of AI to enhance the administration of protected areas (Fig. 3). AI algorithms have greatly improved traditional tactics, such as patrol routes and anti-poaching operations, by accurately predicting poaching hotspots using historical data and real-time inputs (Cronin et al. 2021; Chisom et al. 2024). Predictive modeling in African national parks assists rangers by predicting the probability of poaching episodes, enabling proactive measures that are consistent with conventional patrolling methods (Sarkar et al. 2022).

Additionally, AI technology can provide efficient assistance for the community-based conservation programs, which are a conventional method that incorporates indigenous knowledge and practices related to resource management (Ruiz-Mallén and Corbera 2013). The effective monitoring of environmental changes and animal trends can be achieved by communities through the utilization of AI-driven data analytics (Chisom et al. 2024). The utilization of AI technologies in Nepal has been implemented to analyze photos and data obtained from community-managed forests (Paudel and Sah 2015). This utilization aids local stakeholders in making well-informed decisions regarding sustainable practices and intervention measures.

Furthermore, AI significantly improves the efficacy of habitat restoration initiatives that have conventionally relied on manual monitoring and labor-intensive procedures (Haq et al. 2024). The utilization of AI-powered drones and remote sensing technologies has become prevalent in the mapping and monitoring of restoration work across extensive regions. These technologies offer high-resolution data that effectively inform reforestation initiatives and facilitate biodiversity evaluations (Lock et al. 2022). The utilization of this technology was successfully implemented in the Amazon Rainforest, where AI-driven



drones evaluate the results of reforestation efforts and assist in optimizing planting techniques to guarantee the restoration of the ecological system (Hodel 2023).

Challenges and limitations of implementing AI in conservation

Although the incorporation of AI in conservation has demonstrated encouraging progress, it also poses several obstacles and constraints that require thorough assessment (Sarkar et al. 2006). A notable obstacle is to the caliber and accessibility of data necessary for the training of AI models (Hu and Xu 2023). In certain biodiversity hotspots, such as tropical forests, the limited availability of data can hinder the efficacy of AI applications. The efficacy of species identification and behavioral prediction models is contingent upon the size and caliber of the input data, which may be limited or insufficient in biodiversity areas (Sarkar and Chapman 2021).

Another constraint pertains to the expenses and intricacy associated with implementing and up keeping AI technologies, particularly in distant or underprivileged regions. The establishment and continuous upkeep of sensors, drones, and other AI-powered devices can pose significant financial and technological challenges, necessitating specific expertise that may not be readily accessible within the local context (Bibri et al. 2024). Long-term field studies aided by AI in the conservation of marked ungulates encounter logistical problems, such as the requirement for ongoing finance and human resources (Eikelboom 2021).

In addition, the dependence on AI might give rise to ethical considerations, namely pertaining to monitoring and privacy (Kochupillai et al. 2022). The utilization of AI-powered cameras and tracking equipment in wildlife monitoring gives rise to concerns regarding the effects on animal welfare and the ethical implications of continuous surveillance (Basu and Nath 2024). The integration of AI in conservation efforts necessitates a careful equilibrium between technological encroachment and the preservation of wildlife's natural behavior and environment.

Moreover, AI models can occasionally function as opaque systems, with decision-making procedures that lack transparency or comprehensibility for conservation practitioners (Wünscher and Engel 2012). The absence of transparency can impede the establishment of confidence and acceptance of AI technologies, as it is imperative that decisions pertaining to species conservation and habitat management are comprehensible to a wide range of individuals, including those lacking technical proficiency (Raihan 2023).

AI technologies have the potential to worsen inequities if they are not implemented with careful consideration (Dauvergne 2020). In specific circumstances, the use of high-tech solutions has the potential to redirect financial resources away from conventional conservation approaches that are seen more culturally suitable or environmentally sustainable (Lama and Sattar 2002). This phenomenon has the potential to result in a disparity in technological availability, wherein more affluent conservation initiatives possess advanced tools, while economically disadvantaged regions fall behind.

Ethical considerations and long-term sustainability of AI-driven conservation

The ethical implications and enduring viability of AI-driven conservation are contingent upon various interconnected aspects that involve the integration of ethical theory and practical implementation (Kochupillai et al. 2022). Environmental ethics must fundamentally influence the implementation of AI in conservation initiatives, guaranteeing that technology progress does not overshadow ecological or social principles (Nandutu et al. 2023).

Ethical AI in environmental situations frequently entails making intricate choices that strike a balance between technology capabilities, ecological consequences, and human welfare.

The environmental impact of AI technologies themselves is a significant concern when considering the sustainability of AI-driven conservation efforts (Akter 2024). AI systems, especially those that utilize huge data centers and substantial computational resources, make a substantial contribution to carbon emissions (Gaur et al. 2023). Therefore, it is imperative to create 'sustainable AI' that takes into account the ecological expenses associated with technology (Akter 2024). The utilization of advancements in AI hardware and software to mitigate energy consumption, as well as the integration of renewable energy sources to support power-intensive AI operations. Furthermore, ethical AI must tackle concerns regarding prejudice, unfair treatment, and the possibility of worsening disparities (Ozmen Garibay et al. 2023). Within the realm of conservation, it is imperative to guarantee that AI does not unintentionally exhibit bias towards some species or ecological systems, while disregarding others (Ditria et al. 2022). Additionally, AI must not disregard the perspectives and entitlements of local communities impacted by conservation initiatives (Taylor et al. 2022). For example, the utilization of AI in the administration of animal territories and parks should not result in the implementation of exclusionary measures that isolate indigenous communities who may depend on these lands for their sustenance. An additional ethical consideration arises from the potential danger of over dependence on AI solutions, which could result in a disengagement from conventional conservation techniques and knowledge systems. It is necessary to adopt a well-rounded strategy that utilizes AI to augment, rather than supplant, conservation initiatives headed by humans. This entails actively involving stakeholders at every level to guarantee the responsible and transparent utilization of AI tools.

Conclusion

This study has presented the interactions between classical and modern methodologies in the field of biodiversity conservation, emphasizing the specific advantages and obstacles that each approach brings in the pursuit of safeguarding the biological diversity of our world. Classical approaches, which are based on well-established techniques such as the establishment of protected areas and the implementation of species-specific management, provide a robust framework of proven strategies. However, these methods frequently lack the adaptability required to effectively respond to swift environmental transformations and intricate socio-economic circumstances. In contrast, contemporary methodologies, driven by advancements in AI and associated technologies, introduce a sense of dynamism and scalability to conservation endeavors. These methodologies offer tools capable of analyzing extensive datasets and automating intricate operations with unparalleled accuracy. The incorporation of AI into conservation methods signifies a revolutionary change that has the capacity to improve the extent and efficiency of conservation efforts greatly. AI powered solutions enhance the process of gathering and examining data, while also facilitating live monitoring and predictive analytics. This empowers conservationists to anticipate and address possible hazards before they materialize. However, further research should be conducted to investigate the potential synergies between classical and modern AI techniques. By adopting a comprehensive strategy that acknowledges the value of conventional conservation techniques as well as the capabilities of contemporary technologies, it is possible to

envision the simultaneous attainment of ecological preservation and sustainable development objectives.

Author contributions F.U. and S.S. co-wrote the main manuscript text. Y.-C.X. prepared and revised the manuscript. All authors reviewed the manuscript.

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Declarations

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